2.10.54

EE25BTECH11025 - Ganachari Vishwambhar

Question:
let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be unit vectors such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$. Which of the following are correct?

1) $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} = \mathbf{0}$
2) $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \neq \mathbf{0}$
3) $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{a} \times \mathbf{c} \neq \mathbf{0}$
4) $\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}$ are mutually perpendicular.

Solution:

Given:

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = 0 \tag{1}$$
$$\mathbf{c} = -(\mathbf{a} + \mathbf{b}) \tag{2}$$
$$\mathbf{b} = -(\mathbf{a} + \mathbf{c}) \tag{3}$$
$$\mathbf{a} = -(\mathbf{b} + \mathbf{c}) \tag{4}$$

Since $\mathbf{c}$ is a unit vector:

$$\|\mathbf{c}\| = 1 \tag{5}$$
$$\tag{6}$$

From(2):

$$1 = \|\mathbf{c}\|^2 = (\mathbf{a} + \mathbf{b})^\top (\mathbf{a} + \mathbf{b}) \tag{7}$$
$$1 = \mathbf{a}^\top\mathbf{a} + \mathbf{b}^\top\mathbf{b} + 2\mathbf{a}^\top\mathbf{b} \tag{8}$$
$$1 = 1 + 1 + 2\mathbf{a}^\top\mathbf{b} \tag{9}$$
$$\mathbf{a}^\top\mathbf{b} = \frac{-1}{2} \tag{10}$$

Similarly,

$$\mathbf{b}^\top\mathbf{c} = \frac{-1}{2} \tag{11}$$
$$\mathbf{c}^\top\mathbf{a} = \frac{-1}{2} \tag{12}$$

$\therefore$Angle between the vectors is neither $0°$ nor $180°$:

$$\mathbf{a} \times \mathbf{b} \neq 0 \tag{13}$$
$$\mathbf{b} \times \mathbf{c} \neq 0 \tag{14}$$
$$\mathbf{c} \times \mathbf{a} \neq 0 \tag{15}$$

Proving $\mathbf{b} \times \mathbf{c} = \mathbf{a} \times \mathbf{b}$.

From(2):

$$\mathbf{b} \times \mathbf{c} = \mathbf{b} \times (-(\mathbf{a} + \mathbf{b})) \tag{16}$$

$$\mathbf{b} \times \mathbf{c} = -(\mathbf{b} \times \mathbf{a}) - (\mathbf{b} \times \mathbf{b}) \tag{17}$$

$$\mathbf{b} \times \mathbf{c} = \mathbf{a} \times \mathbf{b} \tag{18}$$

$$\because \mathbf{b} \times \mathbf{b} = 0 \tag{19}$$

Similarly,

$$\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c}. \tag{20}$$

$$\mathbf{c} \times \mathbf{a} = \mathbf{a} \times \mathbf{b}. \tag{21}$$

From (13, 14, 15, 18, 20, 21), we can conclude that:

$$\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \neq 0 \tag{22}$$

Hence, option (3) is correct.

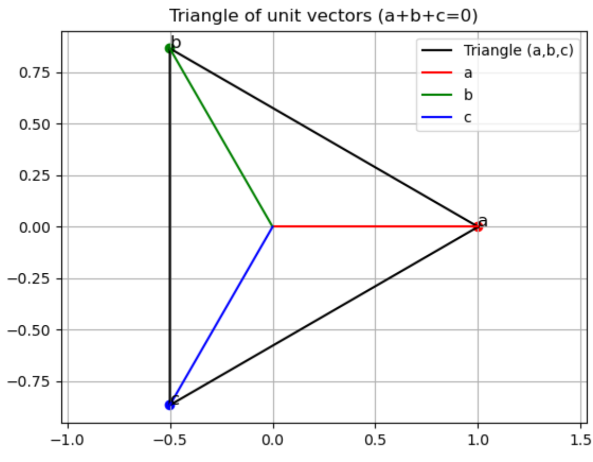

Fig. 1: Plot of the given line, point and reflected point